A11.3 Catalysts and Reaction Rate

Definition • alternative pathway • activation energy • Maxwell–Boltzmann distribution

Learning outcomes

- Define a catalyst accurately.
- Use reaction evidence to decide whether a substance is acting as a catalyst.
- Explain how a catalyst increases reaction rate by providing an alternative pathway with lower activation energy.
- Interpret the effect of a catalyst using a Maxwell–Boltzmann distribution.
- Explain why lowering activation energy increases the proportion of successful collisions.

Starter: collision theory retrieval

- Define activation energy.
- State the two requirements for a successful collision.
- At a fixed temperature, what determines the distribution of particle energies?
- Predict how a reaction could be made faster without increasing temperature.

Starter answers

- Activation energy is the minimum collision energy required for a successful reaction.
- Particles must collide with sufficient energy and the correct spatial orientation.
- At a fixed temperature, particles have a distribution of kinetic energies represented by a Maxwell–Boltzmann distribution.
- One method is to use a catalyst.

A11.3.1 Definition of a catalyst

- A catalyst is a substance that increases the rate of a chemical reaction without itself being permanently chemically changed.
- The catalyst participates in the reaction mechanism but is regenerated overall.
- It is therefore not used up permanently during the reaction.
- A correct definition must include both increased rate and no permanent chemical change.

What a catalyst does not mean

- A catalyst does not increase the amount of reactant present.
- A catalyst does not increase the temperature of the reaction mixture by definition.
- A catalyst is not simply 'a substance that makes a reaction happen'.
- The key effect is a faster reaction through a different reaction pathway.

Catalysts provide an alternative pathway

- A catalyst provides an alternative reaction pathway.
- This pathway has a lower activation energy than the uncatalysed pathway.
- At the same temperature, the particles have the same overall energy distribution as before.
- Because the activation-energy threshold is lower, more particles have sufficient energy to react.

Link to collision theory

- Without catalyst: only collisions with energy ≥ the higher Ea can satisfy the energy requirement.
- With catalyst: the lower Ea means a greater proportion of collisions have sufficient energy.
- Correct spatial orientation is still required.
- More successful collisions per unit time → increased rate of reaction.

Activity: Catalyst Evidence Analysis

Teacher action

- Present reaction-time or rate data with and without a suspected catalyst.
- Model how evidence supports both parts of the definition.

Student action

- Compare reaction rates.
- Decide whether the added substance is consistent with a catalyst.
- Justify using increased rate and the requirement that it is not permanently chemically changed.

Catalyst Evidence Analysis: feedback

- A shorter reaction time or greater measured rate supports the claim that the substance increases reaction rate.
- Evidence that the substance is recovered/regenerated or is not permanently chemically changed supports the second part of the definition.
- Rate evidence alone does not establish the complete definition.

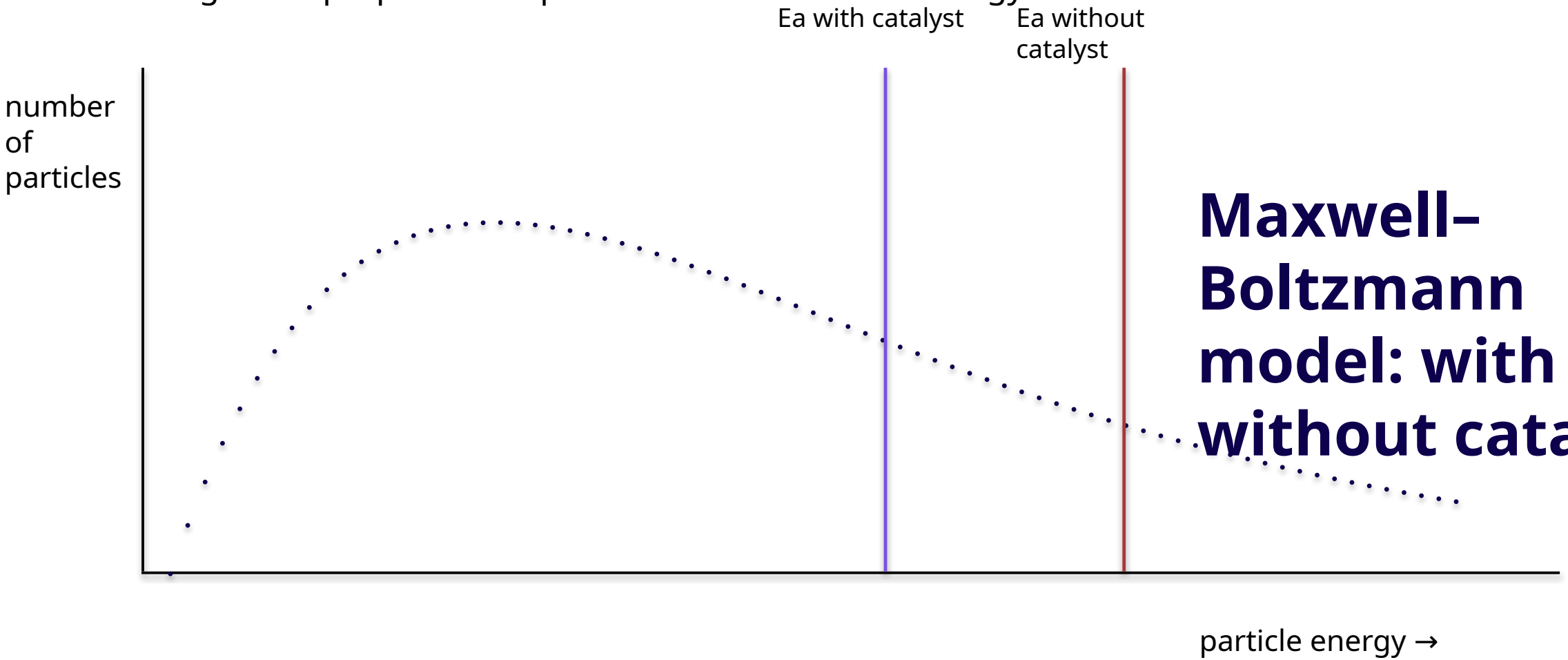
Worked evidence example

- Reaction without substance X: 120 s to reach the stated endpoint.
- Reaction with substance X: 45 s to reach the same endpoint.
- Substance X is recovered after the reaction with no permanent chemical change.
- Conclusion: the evidence is consistent with X acting as a catalyst.

A11.3.2 Maxwell–Boltzmann: catalyst effect

- A catalyst does not need a new Maxwell–Boltzmann curve at the same temperature.
- The particle-energy distribution is unchanged because temperature is unchanged.
- What changes is the activation-energy threshold.
- The catalysed pathway has a lower Ea.

- One energy distribution represents the particles at the fixed temperature.
- The catalysed Ea is positioned to the left of the uncatalysed Ea.
- The area to the right of the catalysed Ea is larger.
- Therefore a greater proportion of particles have sufficient energy to react.



Maxwell–Boltzmann model: with and without catalyst

Reading the graph correctly

- Area to the right of an Ea line represents the proportion of particles with energy equal to or greater than that activation energy.
- Lowering Ea increases this area even though the distribution itself is unchanged.
- This means a greater proportion of collisions can satisfy the energy requirement.
- The frequency of successful collisions therefore increases.

Catalyst versus temperature: do not confuse them

- Increasing temperature changes the Maxwell–Boltzmann distribution.
- Adding a catalyst at the same temperature does not change that distribution.
- A catalyst lowers the activation energy by providing an alternative pathway.
- Both can increase the proportion of particles able to react, but they do so in different ways.

Activity: Catalyst and Maxwell–Boltzmann Analysis

Teacher action

- Model a fixed distribution with two Ea lines.
- Ask students to identify the catalysed and uncatalysed thresholds before discussing area.

Student action

- Identify which Ea corresponds to the catalysed pathway.
- Compare the areas to the right of each Ea.
- Explain why the catalysed reaction is faster using collision theory.

Maxwell–Boltzmann Analysis: feedback

- The lower Ea corresponds to the catalysed pathway.
- The area to the right of the lower Ea is larger.
- A greater proportion of particles therefore have sufficient energy for successful collisions.
- More successful collisions per unit time increases reaction rate.

Common misconceptions

- ‘A catalyst gives particles more energy.’ Incorrect: at the same temperature the energy distribution is unchanged.
- ‘A catalyst lowers the energy of every particle.’ Incorrect: it lowers the activation-energy requirement.
- ‘A catalyst is used up.’ Incorrect: it is not permanently chemically changed overall.
- ‘A catalyst changes the reaction rate by changing temperature.’ Incorrect: its defining role is an alternative lower-Ea pathway.

Independent: Catalyst Identification Challenge

Teacher action

- Provide unfamiliar reaction scenarios including genuine catalysts and distractors.
- Require students to apply the full definition.

Student action

- Identify whether each substance is acting as a catalyst.
- Use rate evidence and information about permanent chemical change.
- Reject cases where only one part of the definition is satisfied.

Independent: Catalyst Graph Challenge

Teacher action

- Provide Maxwell–Boltzmann diagrams with alternative Ea lines.
- Do not alter temperature unless explicitly stated.

Student action

- Identify catalysed and uncatalysed Ea.
- Shade/describe the extra proportion of particles able to react.
- Explain the increased rate using lower Ea and successful collisions.

Guided application

- A reaction has Ea = 85 kJ mol^{-1} without a catalyst and Ea = 52 kJ mol^{-1} with a catalyst. Temperature is unchanged.
- 1. Which activation-energy line lies further left on a Maxwell–Boltzmann graph?
- 2. Does the shape of the distribution change?
- 3. Which pathway has the larger proportion of particles with $E \geq Ea$?
- 4. Explain the effect on reaction rate.

Guided application: answers

- The 52 kJ mol^{-1} catalysed Ea lies further left.
- No. At unchanged temperature the Maxwell–Boltzmann distribution is unchanged.
- The catalysed pathway has the larger proportion of particles with $E \geq Ea$.
- More collisions have sufficient energy, so successful collisions occur more frequently and rate increases.

Error Detective

- Student A: 'A catalyst is any substance that speeds up a reaction.'
- Student B: 'The catalyst raises the Maxwell–Boltzmann curve.'
- Student C: 'The catalyst increases the kinetic energy of the particles.'
- Student D: 'The catalyst lowers activation energy by providing an alternative pathway, so more collisions can be successful.'
- Identify and correct the inaccurate statements.

Error Detective: answers

- A: incomplete: a catalyst increases rate without itself being permanently chemically changed.
- B: incorrect: at the same temperature the distribution is unchanged; the Ea threshold is lower.
- C: incorrect: particle kinetic-energy distribution is unchanged at the same temperature.
- D: correct.

Exam style application

- (a) Define a catalyst. [2]
- (b) Explain how a catalyst increases the rate of a reaction in terms of activation energy. [2]
- (c) A Maxwell–Boltzmann distribution is drawn at one temperature. Explain how the graph should show catalysed and uncatalysed pathways. [3]
- (d) Explain why the catalysed pathway produces a greater proportion of successful collisions. [2]

Exam style application: answers

- (a) Substance that increases the rate of a chemical reaction (1) without itself being permanently chemically changed (1).
- (b) It provides an alternative reaction pathway (1) with lower activation energy (1).
- (c) Use the same distribution at the same temperature (1); show a lower Ea for the catalysed pathway (1); the area to the right of the catalysed Ea is larger (1).
- (d) A greater proportion of particles have E ≥ the lower Ea (1), so more collisions satisfy the energy requirement and can be successful (1).

Final knowledge check

- Give the complete definition of a catalyst.
- State what happens to activation energy when a catalyst provides an alternative pathway.
- At constant temperature, does the Maxwell–Boltzmann distribution change when a catalyst is added?
- What happens to the area representing particles with $E \geq Ea$?
- Explain why reaction rate increases.
- State one misconception about catalysts and correct it.